WORKED SOLUTION

STEP II 2011, Question 10

Mechanics • Projectiles and Impulse

STEP II 2011, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q10

The key idea

Because R is always vertically below P, the two particles share the same horizontal velocity throughout, so the string is vertical and only the vertical motion matters. The string goes taut when P is exactly $\frac{1}{2}H$ above the plane; the jerk is then a vertical impulse that shares P's upward momentum with R. After that the two particles have identical velocities, so they fly together as one projectile until R lands.

Solution

Maximum height

Model the particle as a point under gravity alone. The vertical component of velocity is $u\sin\theta - gt$, which is zero at the top. Using $v^2 = u^2 + 2as$ vertically, with final vertical speed 0,

$$0 = u^2\sin^2\theta - 2gH \qquad \Longrightarrow \qquad H = \frac{u^2\sin^2\theta}{2g}.$$

It is worth noting for later that this gives $u\sin\theta = \sqrt{2gH}$.

When the string becomes taut

Take O as origin, x horizontal and y vertically up. The particle P has horizontal velocity $u\cos\theta$ for all time. For R to stay vertically below P, it must be projected with speed $u\cos\theta$; the plane is smooth, so R then slides at this constant speed. Hence the string is always vertical, and the distance PR is simply the height of P:

$$y_P = u\sin\theta\, t - \tfrac{1}{2}gt^2 = \sqrt{2gH}\, t - \tfrac{1}{2}gt^2.$$

The string becomes taut when y_P first reaches $\frac{1}{2}H$ (which it does, since P rises to $H > \frac{1}{2}H$):

$$\tfrac{1}{2}gt^2 - \sqrt{2gH}\,t + \tfrac{1}{2}H = 0 \quad \Longrightarrow \quad t = \frac{\sqrt{2gH} \pm \sqrt{2gH - gH}}{g} = (\sqrt{2} \pm 1)\sqrt{\frac{H}{g}}.$$

The smaller root is the first time, so the string becomes taut after

$$t_1 = (\sqrt{2} - 1)\sqrt{H/g},$$

as required. (The larger root is when P would come back down through height $\frac{1}{2}H$.)

The jerk

At t_1 the vertical velocity of P is

$$u\sin\theta - gt_1 = \sqrt{2gH} - (\sqrt{2} - 1)\sqrt{gH} = \sqrt{gH},$$

upwards, while R has no vertical velocity. Both have horizontal velocity $u\cos\theta$.

Ask yourself Which impulses act at the instant the string jerks, and in which direction?

The string is vertical, so the tension impulse J is vertical: downwards on P, upwards on R. The plane is smooth and can only push up on R; since R is pulled upwards off the plane, the plane exerts no impulse. So horizontal velocities are unchanged, and vertically the momentum of the pair is conserved. The string is inextensible and stays taut, so just afterwards P and R have the same vertical velocity w. With equal masses m,

$$m\sqrt{gH} + m \cdot 0 = 2mw \qquad \Longrightarrow \qquad w = \tfrac{1}{2}\sqrt{gH}.$$

The distance D

After the jerk, P and R have the same velocity and the same acceleration g downwards if the tension is zero. So they move as two identical projectiles a fixed distance $\frac{1}{2}H$ apart: the string stays exactly taut with zero tension, which is consistent. R starts at height 0 with upward velocity w, so it returns to the plane after

$$t_2 = \frac{2w}{g} = \sqrt{\frac{H}{g}}.$$

The whole time of R's motion is therefore

$$t_1 + t_2 = \sqrt{2}\sqrt{\frac{H}{g}} = \sqrt{\frac{2H}{g}} = \sqrt{\frac{u^2\sin^2\theta}{g^2}} = \frac{u\sin\theta}{g},$$

and throughout it R moves horizontally at $u\cos\theta$. Hence

$$D = u\cos\theta \cdot \frac{u\sin\theta}{g} = \frac{u^2\sin\theta\cos\theta}{g} = \frac{u^2\sin 2\theta}{2g}.$$

This is exactly half the range of P on its own: R lands at the moment P, without a string, would have been at the top of its path.

The case $D = H$

Setting

$$\frac{u^2 \sin\theta\cos\theta}{g} = \frac{u^2 \sin^2\theta}{2g}$$

and dividing by $u^2 \sin\theta/g$ (non-zero, as $0 < \theta < \frac{1}{2}\pi$ for the particle to rise) gives $\cos\theta = \frac{1}{2}\sin\theta$, so

$$\tan\theta = 2.$$

Checks

- The two roots for t multiply to H/g and sum to $2\sqrt{2H/g}$, the time for P to reach the top and come back to the plane, as they should by symmetry.
- A numerical simulation (stepping P's flight until it is $\frac{1}{2}H$ up, applying the jerk, then stepping again until R lands) for random u and θ matched t_1 and D; for example $u = 18.11$, $\theta = 0.711$, $g = 9.8$ gives $D \approx 16.543$ both ways.
- With $\tan\theta = 2$, $u = 10$, $g = 9.8$: $D = H \approx 4.082$.

Common slips

- Taking the larger root $(\sqrt{2}+1)\sqrt{H/g}$, which is when P comes back down through height $\frac{1}{2}H$.
- Forgetting the time t_1 that R spends sliding before the jerk, and computing D from the flight after the jerk alone.
- Using energy conservation at the jerk: kinetic energy is lost when an inextensible string jerks taut; only momentum is conserved.

Beyond STEP

After the jerk the pair behaves exactly like a single particle at its centre of mass: internal forces (here the tension) never change the motion of the centre of mass of a system. This is the starting point for the mechanics of systems of particles and rigid bodies in a first-year course, where impulsive constraints like this one are handled the same way. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

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